Rare network failures via saturated entailment: proof candidate

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Status. This is an internally derived, unreviewed proof candidate for a polynomial-time relative approximation scheme for the probability of an odd-signed cycle in an independently retained signed graph. It includes a spectral-gap argument, an annealing and bit-cost contract, and a planar two-terminal failure corollary. Originality is not established. There is no external or formal certification and no practical speed benchmark.

The candidate changes the marked state space: an odd cycle anywhere makes every pair label feasible. This saturation is essential to the proof. The argument does not establish rapid mixing for an unsaturated affine-fiber chain. The primary prior framework is Feng, Fu and Guo, *Approximating two-terminal network reliability*, arXiv:2608.02523, especially its root-layer flow and comparison lemmas [1].

1 Model and candidate result

Let $G=(V,E)$ be a finite undirected signed multigraph, with $n=|V|$, $m=|E|$, signs $\sigma(e)\in F_2$, and a random retained edge set F drawn from the product law

$$\mu(F)=\prod_{e\in F} p_e \prod_{e\notin F}(1-p_e).$$

The input probabilities are supplied rationals in binary. Write B for their total encoding length. Initially take $0<p_e<1$ and $m\geq 1$; Section 7 treats endpoints and loops. Let O be the event that F contains a cycle whose edge signs sum to one, and let $U=\mu(O)$. A balanced full graph has $U=0$, detected by signed graph search. Assume otherwise that $U>0$.

Use labels

$$L=\{0,o\}\cup\{(u,v,s):u<v,s\in F_2\},\ N=2+2\binom{n}{2}.$$

The labels represent the zero syndrome, the zero-boundary odd syndrome, and two-boundary signed syndromes. Define

$$E_0=2^E,\ E_o=O,$$

$$E_{(u,v,s)}=O\cup\{F:\text{a simple } u-v \text{ path in } F \text{ has sign } s\}.$$

An odd cycle in any component makes every label feasible, including labels with endpoints in different components. Equivalently, the equations $x_u+x_v=\sigma(e)$ for retained edges logically entail the label equation; an inconsistent system entails every equation. On a balanced edge set, a feasible pair is connected and has its uniquely determined path sign.

Put $q_a=\mu(E_a)$. Then $q_0=1$ and $U\leq q_a\leq 1$. Each event is increasing and can be tested in $O(n+m)$ graph operations. Regard labels as vectors in $F_2^V\times F_2$, and write $A_e=(\partial e,\sigma(e))$. Whenever the indicated labels are valid, entailment and Harris correlation give

$$E_a\cap E_b\subseteq E_{a+b},\ q_{a+b}\geq q_a q_b. \quad (1)$$

If $b = a + A_e$, the same reasoning gives

$$E_a \cap \{ e \in F \} \subseteq E_b, q_b \geq p_e q_a, q_a \geq p_e q_b. \quad (2)$$

Candidate theorem. The lazy chain defined below, with ideal weights, satisfies

$$gap(K) \geq \frac{1}{96 N m(m+1)}. \quad (3)$$

Weights within a factor four of $1/q_a$ lose at most a factor 4096 in this bound. Together with Section 6, this yields the candidate FPRAS: for $0 < \varepsilon, \delta < 1$, a relative-$\varepsilon$ estimate of U with success probability at least $1 - \delta$, using time polynomial in $n, m, B, \varepsilon^{-1}, \log(\delta^{-1})$. Sections 2–6 are the proposed proof, subject to the status above.

2 Exact marked chain

The state space consists of (a, F) with $F \in E_a$. Its ideal law and lazy kernel are

$$\pi(a, F) = \frac{\mu(F)}{N q_a}, K = \frac{1}{2} I + \frac{1}{6} K_{edge} + \frac{1}{6} K_{walk} + \frac{1}{6} K_{hub}.$$

Physical refresh. K_{edge} chooses an edge uniformly and heat-bath refreshes its bit conditional on the current label staying feasible. If both values are feasible, the refreshed bit is Bernoulli(p_e); otherwise monotonicity forces it to one.

Label walk. K_{walk} chooses an edge uniformly and one of two modes uniformly. It proposes $b = a + A_e$, rejecting an invalid label. Toggle mode also flips the physical bit of e. Keep mode leaves the physical set fixed and is permitted only when e is present. Infeasible proposals are rejected; feasible proposals use the Metropolis–Hastings acceptance probability for π. Each edge/mode proposal is a symmetric involution. Edge identity and mode remain explicit channel tags, so parallel edges incur no extra ambiguity factor.

Odd-sector hub. K_{hub} chooses $\ell \in L$ uniformly. If $F \in O$ and the current label is one endpoint of the star edge $\ell \leftrightarrow o$, it proposes the other endpoint; otherwise it stays put. It uses Metropolis–Hastings. For $\ell \neq o$, acceptance from (ℓ, F) to (o, F) is one because $q_\ell \geq U$. The proposal probability is $1/N$.

These kernels are reversible. With implementable positive weights c_a, replace $1/q_a$ by c_a throughout the target law and acceptance ratios. Feasibility testing uses graph search; no conditional rare-event sampling oracle is assumed.

3 Routes to a product-law anchor

The anchor is the whole zero-label sector, whose conditional edge law is μ. Route each source (a, F) to $(0, F')$ with F' independently drawn from μ.

Balanced pair sources. For balanced $F \in E_{(u,v,s)}$, run deterministic BFS from u until v is discovered, with fixed adjacency order and each physical edge inspected once. Append the uninspected edges in fixed order. The resulting permutation $e_1, \ldots, e_m$ has a simple discovery path C from u to v of sign s. The inspection indices increase along that path. For $a = 0$ and arbitrary F, use a fixed permutation and $C = \varnothing$.

Odd sources. For label o, run a deterministic signed DFS forest with fixed root and adjacency orders. Present discovery edges set vertex parities. Stop at the first present edge inconsistent with its endpoints' assigned parities, then append uninspected edges in fixed order. In undirected DFS, this closing edge joins a vertex to an active ancestor; its fundamental cycle C is odd. The tree edges on its ancestor-to-descendant chain have increasing inspection indices and precede the closing edge. A negative loop is the one-edge version of this witness.

For either base family, set

$$H^k = F \text{ on } \{e_1, \dots, e_k\}, F' \text{ on } \{e_{k+1}, \dots, e_m\},$$

$$b_k = \sum_{e \in C \cap \{e_1, \dots, e_k\}} A_e.$$

Thus $(b_m, H^m) = (a, F)$ and $(b_0, H^0) = (0, F')$. Every b_k is a valid label. Its remaining certificate lies in H^k, so each state is feasible. For an odd cycle, reverse inspection first removes the closing edge, leaves a pair label, peels the tree path, and ends at zero. At every point there are at most two boundary defects.

Scan $k = m, m-1, \dots, 1$. If $e_k \notin C$, refresh its bit by a physical move when the source and target bits differ. If $e_k \in C$, it is currently present: use a label-walk move to change b_k to b_{k-1}, keeping or deleting the edge according to its target bit in F'. Omit trivial transitions. Each base route has length at most m.

A pair source with odd F first uses the hub move to (o, F) and then the odd base route. Its length is at most $m+1$. Loop-erasing repeated states only reduces route lengths and loads. Balanced-source families are analyzed separately; odd-source demands are merged before encoding. The replay procedure therefore need not infer a nonlocal balanced-versus-odd branch.

4 Replay and complementary encoding

Fix a directed physical transition channel η, its base family and source label a. Suppose the inspected edge is $e^* = e_k$, and the before and after states are $(b, B) = (b_k, H^k)$ and $(b', B') = (b_{k-1}, H^{k-1})$. Define the complementary hybrid

$$W = F' \text{ on } \{e_1, \dots, e_k\}, F \text{ on } \{e_{k+1}, \dots, e_m\}.$$

Encode (F, F') by W with e^* omitted. The transition restores the omitted bit because $W(e^*) = F'(e^*) = B'(e^*)$.

Replay injection. Replay the specified deterministic BFS, DFS or fixed-order search, reading membership from B up to and including inspection of e^* and from W thereafter. The next queried edge depends only on previous answers. Hence this reconstructs the permutation, stopping time, appended suffix and index k, then both F and F'. If the search stopped before e^*, its fixed appended suffix reconstructs the same index. Thus the encoding is injective for fixed channel and source family.

Product weights cancel coordinate by coordinate:

$$\mu(F)\mu(F') = \mu(B)\mu(W). \quad (4)$$

Let $D = C \cap \{e_{k+1}, \dots, e_m\}$ be the erased certificate. It lies in W with e^* absent, and has syndrome $d = a + b$. For a path it is the suffix; for a cycle it is the complementary arc. At endpoints it may be empty or the full odd cycle. Thus d is a valid label and the omitted-edge encoding belongs to E_d.

Write X_{-e} for an edge set restricted to all edges except e. Let μ_{-e^*} denote the product law on all other edges. Injectivity, monotonicity under forcing e^* absent, and (1) applied to $d+b=a$ imply

$$\sum_{\text{encodings}} \mu_{-e^*}\left(W_{-e^*}\right) \le q_d \le \frac{q_a}{q_b}. \quad (5)$$

Equation (5) is the required encoded-mass bound. There is no rank or nullity correction. Its role corresponds to the encoding estimate in the prior flow framework [1].

5 Capacities and spectral gap

A base pair (F, F') carries demand

$$D = \pi(a, F)\,\mu(F') = \frac{\mu(F)\,\mu(F')}{N\,q_a}.$$

For a physical-refresh channel, both bit values are feasible on the route. Its stationary capacity in K is

$$Q = \frac{\mu_{-e}\left(B_{-e}\right)\,p_e\left(1-p_e\right)}{6\,m\,N\,q_b}.$$

By (4),

$$D = 6\,m\,Q\,\frac{q_b}{q_a}\,\mu_{-e}\left(W_{-e}\right).$$

Summing (5) bounds the load of each fixed source-label base family by $6\,m\,Q$.

For a label-walk channel the old physical bit is one. Put $\theta = p_e$ if the new bit is one and $\theta = 1 - p_e$ otherwise. Symmetric Metropolis–Hastings gives

$$Q \ge \frac{\mu_{-e}\left(B_{-e}\right)}{12\,m\,N}\,min\left(\frac{p_e}{q_b}, \frac{\theta}{q_{b'}}\right).$$

Since $q_b \ge p_e\,q_{b'}$ by (2),

$$Q \ge \frac{\mu_{-e}\left(B_{-e}\right)\,p_e\,\theta}{12\,m\,N\,q_b}, D \le 12\,m\,Q\,\frac{q_b}{q_a}\,\mu_{-e}\left(W_{-e}\right).$$

Equation (5) therefore bounds fixed-family load by $12\,m\,Q$. At most N direct balanced-pair and anchor families contribute at most $12\,N\,m\,Q$.

For odd F, the merged demand coefficient at (o, F), including the original odd source, is

$$\pi(o, F) + \sum_{a\,\text{pair}} \pi(a, F) = \frac{\mu(F)}{N}\left(\frac{1}{U} + \sum_{a\,\text{pair}} \frac{1}{q_a}\right) \le N\,\pi(o, F).$$

Hence the entire odd family contributes at most another $12\,N\,m\,Q$ to a physical channel. A hub channel $(a, F) \to (o, F)$ carries total root-directed demand $\pi(a, F)$ and has capacity $\pi(a, F)/(6\,N)$, giving ratio $6\,N$.

All root-directed channel loads are therefore bounded by $R\,Q$, with

$$R = 24\,N\,m, \ell = m + 1.$$

Concatenate two independently routed paths through the same product anchor, reversing the second. The root-layer flow lemma [1, Lemma 17] gives

$$gap(K) \geq \frac{1}{4\,\ell\,R} \geq \frac{1}{96\,N\,m(m+1)}.$$

Channel tagging is valid by summing the channel Dirichlet forms. It also preserves the bound with parallel edges.

6 Annealing sampling and bit costs

Approximate weights and warm starts. If

$$\frac{1}{4\,q_a} \leq c_a \leq \frac{4}{q_a},$$

the normalized law $\pi_c(a, F) \propto c_a \mu(F)$ is within a factor 16 of π. The standard Dirichlet-form comparison, as in [1, Lemma 16], loses at most 4096 in the gap bound. The learning scheme keeps $c_0 = 1$. Starting each draw independently at $(0, F)$ with $F \sim \mu$ has warmness at most $16\,N$ (and exactly N for ideal weights). A sample with total-variation error at most η therefore costs

$$T(\eta) = O\left(N\,m(m+1)\log\frac{N}{\eta}\right) \quad (6)$$

chain transitions, suppressing universal constants including the comparison loss.

Schedule. Every nonempty sector has a certificate with at most *n* edges. If $\beta\,p_e \leq p'_e \leq p_e$, thin a fixed feasible witness to obtain

$$\beta^n q_a(p) \leq q_a(p') \leq q_a(p). \quad (7)$$

Set $\beta = 1 - 1/(2\,max\{1, n\})$ and $p_e(k) = max\{p_e, \beta^k\}$. Start with the known values $q_a(0) = 1$ and stop at the first level L where every coordinate equals its target. Then

$$\beta^n \geq \frac{1}{2},\ L = O\left(1 + n\log\frac{1}{p_{min}}\right) = O(n\,B).$$

No chain is run at the degenerate all-one initial level. Its known weights are already within a factor two at level one, whose probabilities are strictly below one. At later levels, constant-accuracy weights from the previous level remain within a fixed factor by (7).

Absolute probability estimates. Write $Z = \sum_a c_a q_a$ and let r_a be the stationary label probability. Then

$$r_a = \frac{c_a q_a}{Z},\ r_0 = \frac{1}{Z},\ q_a = \frac{r_a}{c_a r_0}. \quad (8)$$

All r_a are $\Theta(1/N)$. Multiplicative concentration of label counts estimates all ratios in (8) to relative accuracy α using $O(N\,\alpha^{-2}\log(N/\zeta))$ independent stationary draws, with failure probability at most ζ. Independent restarted chains with a sufficiently small total-variation budget replace those stationary draws.

Intermediate levels use fixed accuracy, for example $1/10$, then set c_a to the reciprocal of the resulting estimate and retain $c_0 = 1$. Under that accuracy event, the next level's weights remain within the factor-four band. The final level estimates $q_o = U$ with $O(\varepsilon)$ accuracy. Errors do not multiply in a telescoping product: the exact anchor normalizes absolute probabilities at each level.

Allocate concentration failure at most $\delta/2$ over all levels. A conservative total number of restarted draws is

$$M = O\left(N\left(L + \varepsilon^{-2}\right) \log \frac{N(L+1)}{\delta} \right). \quad (9)$$

Use a deterministic upper bound for M to set each draw's total-variation error at most $\delta/(4M)$. Conditional on all preceding successful estimates, the factor-four mixing bound applies. A union bound, or a sequential coupling to independent stationary draws, charges all mixing error at most $\delta/4$. Equation (6) then yields a total of

$$O\left(M N m(m+1) \log \frac{N M}{\delta} \right) \quad (10)$$

chain transitions, plus M product-law initializations. Each transition uses $O(n+m)$ graph operations and rational arithmetic. Thus graph work is the bound (10) multiplied by $O(n+m)$, with $O(M m)$ additional initialization work. This is a conservative asymptotic resource bound, not an implemented large-instance timing result.

Bit model. A witness gives $q_a \geq p_{min}^n$, so reciprocal weights have $O(nB)$ logarithmic magnitude. The schedule numbers β^k have $O(L \log(2n))$ bits. Empirical updates have the form $c_a \leftarrow c_a \hat{r}_0 / \hat{r}_a$; each ratio of integer counts adds $O(\log M)$ numerator and denominator bits. Over L levels all stored weights, probabilities and Metropolis–Hastings ratios have polynomial bit length in the stated inputs. Counts that vanish are included in the concentration-failure event; an implementation can abort that unsuccessful run rather than divide by zero.

Exact rational Bernoulli generation has polynomial expected bit cost. A strict polynomial runtime version can cap its rejection trials, charging a total failure budget at most $\delta/4$ across the bounded number of random choices. Equivalently, polynomial-precision transition simulation can charge aggregate error within that budget. This accounts for random-bit generation and arithmetic without a real-number oracle. For an explicit crude bit-work bound, let $T = T(\delta/(4M))$ and choose upper bounds

$$S = O\left(nB + L\left[\log(2n) + \log(M+1) \right) + \log(Nm+1) \right),$$

$$J = O\left(M(T+m) + NL \right).$$

Here S bounds rational operand length and J bounds transitions, initialization draws and weight updates. Schoolbook arithmetic and at most $O(\log(J/\delta))$ rejection trials per exact random choice give the conservative bit-work bound

$$O\left(J\left[(n+m)\log(2(n+m)) + S^2 \right) \log(J/\delta) \right).$$

The independent Bernoulli input model is an assumption: estimating or validating physical failure probabilities from observations is outside this cost bound.

7 Zero and certain probabilities and loops

Remove edges of probability zero. If the probability-one subgraph is unbalanced, return $U = 1$. Otherwise choose a consistent sign gauge on each of its connected components and contract those components. The remaining random edges stay independent and have probabilities strictly between zero and one; their signs are adjusted by the gauge. Contraction can create parallel edges and loops. If the resulting full graph is balanced, return $U = 0$.

A positive loop never belongs to a witness and can be discarded for the target probability, or independently restored when a complete physical sample is desired. A negative loop has $A_e = (0, 1)$ and is an odd certificate of length one. DFS tests it on inspection; its reverse certificate route goes from o directly to zero. The label walk exchanges 0 and o, or flips a pair's sign. Presence of a negative loop makes every label feasible, so (1), (2), the replay injection and the capacity argument are unchanged. This closes the multigraph and endpoint cases used by the planar corollary.

8 Planar two terminal failure corollary

Let a finite plane graph have terminals s, t and independent physical edge failures with supplied rational probabilities f_e. If the terminals are disconnected even when every edge works, failure probability is one. Otherwise fix any simple primal s–t path P.

Construct the planar dual. Give dual edge e^* sign one exactly when its primal edge belongs to P, and retain it independently with probability f_e. Dual loops and parallel edges are retained. Then

$$\{ s \text{ and } t \text{ are disconnected by failed edges} \} = \{ \text{the retained signed dual contains an odd cycle} \}. \quad (11)$$

To prove the forward direction, failed edges contain a separating cut ∂S. Its dual is Eulerian and has total signature $|\partial S \cap P| \bmod 2 = 1$. A cycle decomposition therefore contains an odd cycle of failed dual edges. Conversely, a simple odd dual cycle crosses P oddly and separates s from t by planar cycle-cut duality. All crossed primal edges are failed, so no working terminal path remains. A dual loop represents a primal bridge and obeys the same argument.

Consequently, the candidate signed-graph FPRAS gives a candidate FPRAS for planar two-terminal unreliability, using polynomial preprocessing and the unchanged failure probabilities. It estimates failure relatively and directly. A supplied planar embedding, or polynomial-time embedding construction, is required. The reduction is elementary; no novelty is asserted. It gives no general nonplanar unreliability result or consequence for general cographic cuts, and it does not cover correlated failures.

9 Finite checks and remaining validation

Two finite checks support debugging of the argument; neither substitutes for the general proof.

The first enumerated **143,528 base paths and 517,912 transition encodings** across all eight gauge-inequivalent signings of K_4, a remote odd triangle, a signed parallel-edge cycle, and a lollipop with a shortcut. The balanced K_4 signing was detected separately. At $p = 1/2$, checks covered prefix feasibility, complementary certificates, omitted-edge injectivity, replay reconstruction and exact root load-to-capacity ratios. Every reported check passed, and the measured ratios stayed below $24\,N\,m$.

A separate implementation checked an odd five-cycle, a remote path plus odd triangle, competing odd triangles and a negative loop at $p = 1/2$, 0.05 and 0.001. It enumerated 157,956 routes across those 12 cases, including $U = 10^{-15}$. Reported maximum load-to-capacity ratios divided by $N\,m$ ranged from approximately 3.82 to 10.10, below the proposed bound 24. These were small-instance route and congestion checks, not a full annealed sampler benchmark or a test of all rational probability assignments.

Independent mathematical review is still needed for the general flow construction, its composition with the comparison lemma and the complete randomized algorithm. The result remains a proof candidate. No claim of external certification, established originality, practical acceleration or coverage of arbitrary binary XOR systems follows from these checks. General XOR systems need not have this polynomial-size family of bounded-defect exploration certificates.

Reference

[1] Weiming Feng, Yucheng Fu and Heng Guo. *Approximating two-terminal network reliability*. arXiv:2608.02523, submitted 3 August 2026. https://arxiv.org/abs/2608.02523. Primary source for the marked-sector, root-layer multicommodity-flow and approximate-weight comparison framework; see especially Lemmas 16–20. The saturated signed-entailment construction above is a separate candidate derivation, whose originality has not been established.